version.

Can we do better? Yes, we can, in at least two different ways: to see how, we will need first to introduce a new algorithm, and then a game-changing programming model.

13.1.3 Canopy Clustering

What if we could run a quick coarse-grained pseudo-clustering, before running any real clustering algorithm, to get an idea of the distribution of data?

Canopy clustering is normally used for this purpose; it groups points into spherical regions (circles in our 2D examples), like k-means, but unlike it, these regions can overlap, and most points are assigned to more than one region.

The canopy clustering algorithm is faster and simpler than k-means, as it runs in a single pass, doesn’t have to compute the centroids for the canopies (spherical pseudo-clusters), and doesn’t compare each point to each centroid; instead, it elects one point in the dataset as the center of each canopy, and adds points around it to the canopy.